

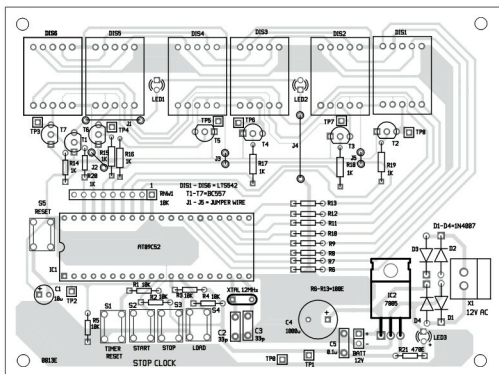


Fig. 4: Component layout for the PCB

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

12 V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C4 and regulated by IC 7805 (IC2). Capacitor C5 by-passes ripples present in the regulated output. LED3 acts as the power indicator. Resistor R21 acts as the current limiter for LED3. A rechargeable battery (BATT.) is used for power backup.

Working of the project is simple.

When power is switched on, it displays '00.00.00,' and both LEDs (LED1 and LED2) are 'off.'

The stop-clock can display minutes, seconds and centi-seconds, and both the LEDs remain 'off' up to one hour. After one hour, the centi-seconds disappear and both the LEDs glow to indicate that the reading is in hours.

The functions of all the four switches are as follows:

Reset (S1): Reset the stop clock at any time (whether the stop clock is running or stopped)

Start (S2): Start the stop clock again

Stop (S3): Stop the stop clock

Load (S4): Record the timings when

the clock is running. The recorded reading can be seen after pressing 'Stop.'

Construction and testing

An actual-size, single-side PCB for real-time clock including the power supply section is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB to minimise assembly errors and save time. Carefully assemble the components and double-check for any overlooked error. Use proper IC base for the microcontroller. Before inserting the ICs, check all the supply voltages.

To verify the circuit for proper functioning, check correct power supply at test point TP0.

Reset signal for the microcontroller can be verified at TP2. Check the display-enable signal at TP3 through TP6 for corresponding display.

Software

Software for the real-time clock is written in 'C' language and compiled using the Keil μ Vision4 compiler. The generated hex code is burnt into the microcontroller by using a suitable programmer. All the switches are regularly scanned. When any switch goes low, corresponding action is performed and displayed. ●

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